

HINTS & SOLUTIONS

Q.1 (B)

Q.2 (B)

* B.P. NF_3 is less than NMe_3 as NMe_3 has more easily polarisable electron cloud.

* $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$
min B.P. max. B.P. (more extant of polarisation)

Q.3 (D)

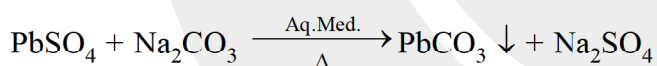
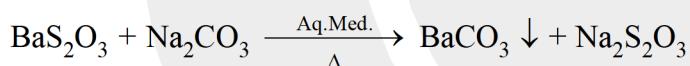
Xe $\mu = 0$ non. polar
 H_2O $\mu \neq 0$ polar
when Xe is dissolved in water

$\delta^- \delta^+$ induced dipole
 $\delta^- \text{H}_2\text{O} \delta^+$ dipole

Q.4 (D)

$(\text{NH}_4)_2\text{SO}_4$, Na_2S and KBr are water soluble while AgCl is insoluble in water, hence for AgCl sodium carbonate extract is formed.

Q.5 (C)

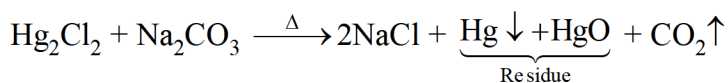


Q.6 (B)

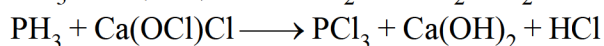
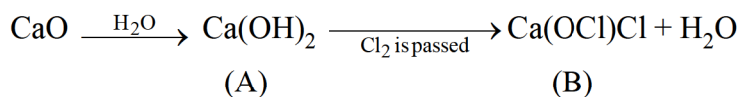
Except NaHCO_3 other sodium salts are water soluble.

Na_2SO_4 is not used, as sulphates of most basic radicals are water soluble.

K_2CO_3 is not used because some acidic radicals like ClO_4^- , $\text{H}_4\text{C}_4\text{O}_6^{2-}$ form insoluble ppt. with K^+ , also fusion temperature of K_2CO_3 is higher than that of Na_2CO_3

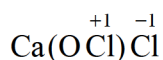


Q.7 (C)



(Reactions are unbalanced)

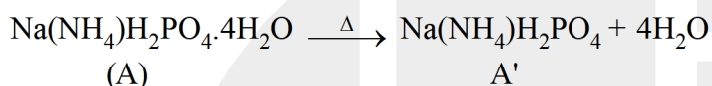
Q.8 (C)



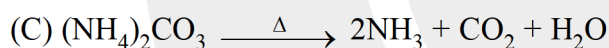
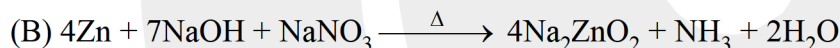
Q.9 (D)

CO_2 or SO_2 or $\text{CO}_2 + \text{SO}_2$ can turn lime water milky.

Q.10 (C)



Q.11 (D)



Q.12 (B)

Q.13 (C)

Q.14 (B)

Q.15 (B)

Q.16 (C)

Q.17 (C)

When size of anion is very large as compared to the size of cation, solubility decreases with increase in size of cation.

Q.18 (B)

- Q.19 (A)
 $\text{Ca(NH)} + 2\text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 + \text{NH}_3 \text{ (g)}$
 (B)
 $2\text{NH}_3 + 3\text{CaOCl}_2 \longrightarrow \text{N}_2 \text{ (g)} + 3\text{CaCl}_2 + 3\text{H}_2\text{O}$
 (B) (C)
 $\text{N}_2 \text{ (g)} + 3\text{Mg} \longrightarrow \text{Mg}_3\text{N}_2$
 (C) (D)
 $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \longrightarrow 3\text{Mg(OH)}_2 + 2\text{NH}_3$
 (D) (B)
- Q.20 (D)
- Q.21 (D)
- Q.22 (B)
 $\text{Ca(OH)}_2 + \text{Na}_2\text{CO}_3 \xrightarrow{\Delta} 2\text{NaOH} + \text{CaCO}_3$
 (A) (B) (C)
 $\text{Ca(OH)}_2 \xrightarrow{\text{CO}_2} \text{CaCO}_3 \downarrow$
 (A) (C)
- Q.23 (A)
 (A) $\text{SiC} \Rightarrow \text{Covalent carbide.}$
 (B) $\text{Be}_2\text{C} + 4\text{H}_2\text{O} \longrightarrow 2\text{Be(OH)}_2 + \text{CH}_4 \uparrow$
 (C) $\text{CaC}_2 + 2\text{H}_2\text{O} \longrightarrow \text{HC} \equiv \text{CH} \uparrow + \text{Ca(OH)}_2$
 (D) $\text{Mg}_2\text{C}_3 + 2\text{H}_2\text{O} \longrightarrow 2\text{Mg(OH)}_2 + \text{H}_3\text{C} - \text{C} \equiv \text{CH} \uparrow$
- Q.24 (C)
 Reactivity of Cu, Ag and Au follows the order $\text{Cu} > \text{Ag} > \text{Au}$.
- Q.26 (D)
 $\text{BeF}_2 + 2\text{NaF} \longrightarrow \text{Na}_2[\text{BeF}_4]$
- Q.27 (C)
- Q.28 (D)
 $\text{Li}^+ > \text{Be}^{+2} \rightarrow \text{Both ions are isoelectronic species}$
 Size of ion decreases
- Q.29 (B)
- Q.30 (B)
 $\rightarrow \text{Na}_2\text{CO}_3$ precipitates out Ca^{++} and Mg^{++} from hard water as insoluble carbonates
 $\rightarrow \text{Calgon; (NaPO}_3)_6$; exchanges Ca^{++} and Mg^{++} from hard water
 $\rightarrow \text{Permutit}$: common type of zeolite, it exchanges Ca^{++} and Mg^{++} from hard water with its Na^+ and also does not allow to pass large sized molecules from it.
- Q.31 (D)

Q.32 (D)

Q.33 (C)

Q.34 (C)

Q.35 (A)

- (A) Uyneco's method (Preparation of H_2)
 $2Al + 2KOH + 2H_2O \rightleftharpoons 2KAlO_2 + 3H_2 \uparrow$ (Very pure)
- (B) Dennis method (Preparation of F_2)
- (C) Deacon's process (Preparation of Cl_2)
- (D) Down cell method (Extraction of Na)

Q.36 (B)

Due to higher lattice energy

Q.37 (C)

Q.38 (C)

Q.39 (A)

Q.40 (B)

Q.41 (B)

Q.42 (D)

Q.43 (C)

Q.44 (B)

Q.45 (D)

Q.46 (A)

Q.47 (A)

Q.48 (B)

Q.49 (A)

Q.50 (B)

Alums are $M_2'(SO_4) M_2''(SO_4)_3 \cdot 24H_2O$

These are soluble in water. Their aqueous solution is acidic both the statements are true but it is not the correct explanation.

Q.51 (A)

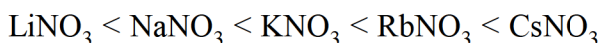
In Alkali metal bicarbonates intermolecular hydrogen bonding is involved and therefore more energy is required to dissociate metal bicarbonate lattice by H_2O .



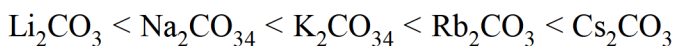
Q.52 (A)

Q.53 (BCD)

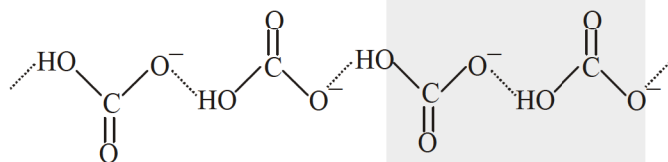
Thermal stability of Nitrates of alkali metals



Metal carbonate solubility



Only NaHCO_3 is insoluble in water due to association of HCO_3^- by intermolecular H-bonding



Q.54 (AB)

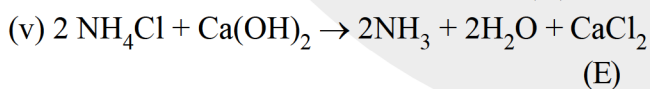
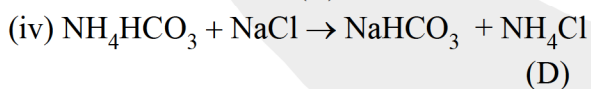
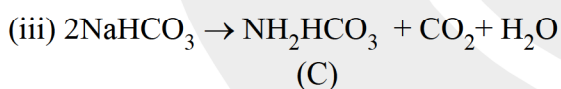
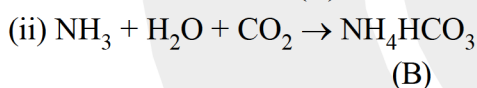
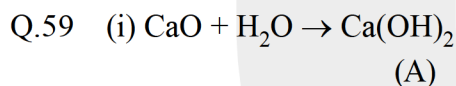
Due to high ionisation energy of Be and Mg

Q.55 (BD)

Q.56 (A) PQRS, (B) PQR (C) PQRS]

Q.57 (A) S (B) R (C) Q (D) T]

Q.58 (3)



Thus, A = Ca(OH)_2 , B = NH_4HCO_3
C = Na_2CO_3 , D = NH_4Cl and E = CaCl_2

Q.60 (8)